Module 8: Cellular Genetics - Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Relate chromosomes to DNA organization.
2. Compare the purpose and products of mitosis and meiosis.
3. Explain how meiosis reduces chromosome number.
4. Describe how crossing over and independent assortment increase variation.
5. Interpret chromosome number errors at an introductory level.

Topics

- Chromosomes and cell cycle
- Mitosis for growth and repair
- Meiosis for gametes
- Variation from recombination and assortment
- Errors and genetic consequences

Key Terms to Know

- **Chromosome** - DNA-protein structure carrying genes.
- **Cell cycle** - Ordered sequence of cell growth, DNA replication, and division.
- **Mitosis** - Nuclear division producing identical daughter nuclei.
- **Meiosis** - Division process producing haploid gametes.
- **Gamete** - Haploid reproductive cell.
- **Crossing over** - Exchange of DNA between homologous chromosomes.
- **Independent assortment** - Random orientation of chromosome pairs in meiosis.
- **Nondisjunction** - Failure of chromosomes to separate correctly.

Core Contents

1. Chromosomes organize DNA so cells can divide accurately.
2. Mitosis produces genetically identical daughter cells for growth and repair.
3. Meiosis produces haploid gametes for sexual reproduction.
4. Crossing over and independent assortment create genetic variation.
5. Nondisjunction and other errors can change chromosome number.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-08_cellular-genetics.md`

Generated Visuals

- Module 08: Cellular Genetics Evidence Map
- Module 08: Cellular Genetics Reasoning Sequence
- Module 08: Cellular Genetics Retrieval and Lab Check